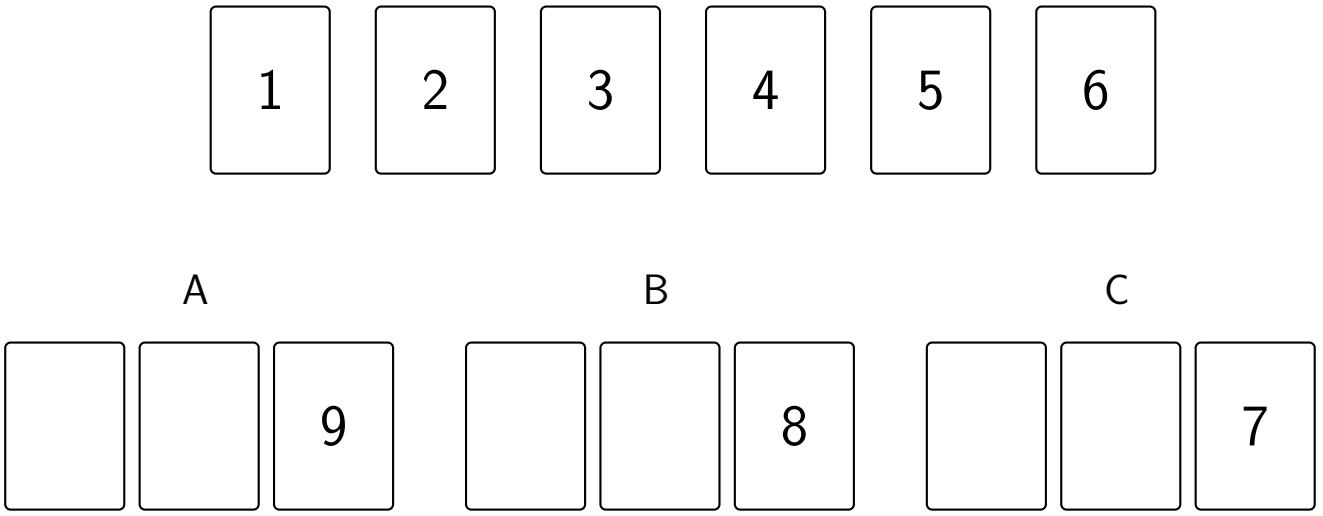
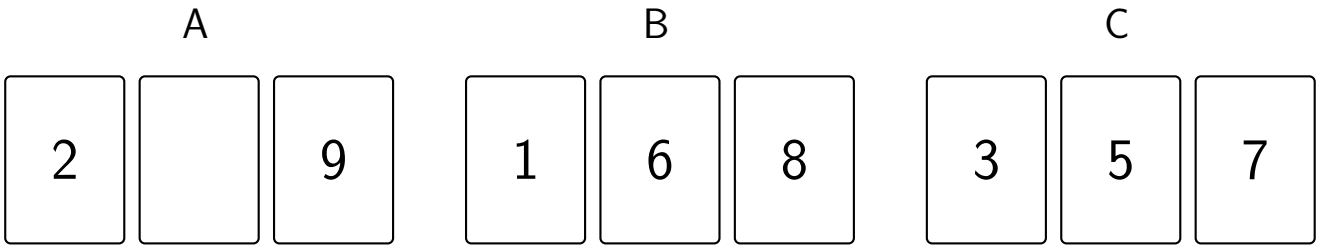
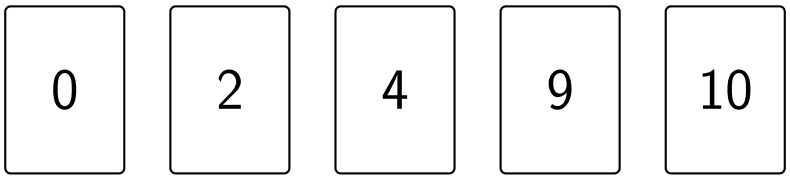
Problem 4: Keep 9 in A, 8 in B, and 7 in C. Place cards 1 to 6 so A wins more pairs than B, B more than C, and C more than A. Find two arrangements different from Problem 2.



Problem 5: Choose 0, 2, 4, 9, or 10 for A's empty card. Find every choice that makes A win more pairs than B, B more than C, and C more than A.





Problem 6: How many possible card pairs does A win against B with the six-card decks, and how many does B win? Then find which deck wins more pairs in each comparison of the four-card decks. Does any winner change from Problem 2?

A

2	2	4
4	9	9

B

1	1	6
6	8	8

A

2	2	4
9		

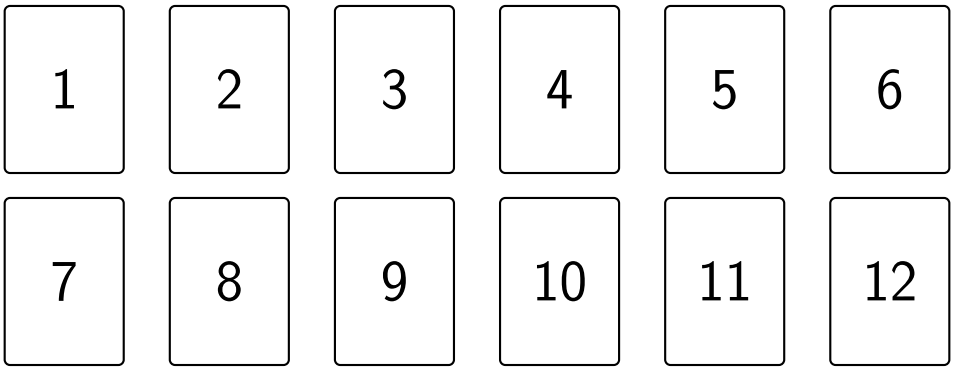
B

1	1	6
8		

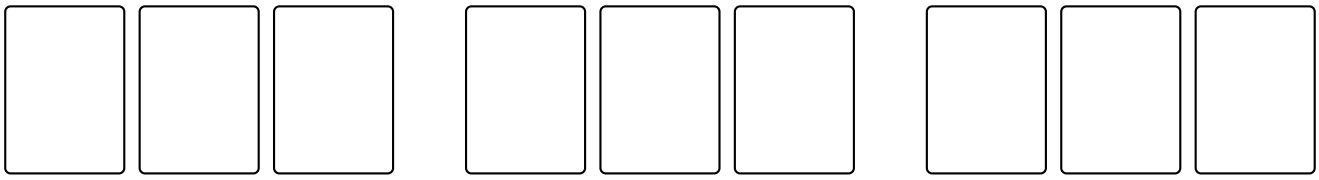
C

3	3	5
7		

Problem 7: Choose nine different cards from 1 to 12 and make three decks of three. Make A win more pairs than B, B more than C, and C more than A, with totals 15 in A, 18 in B, and 21 in C.



A B C



Problem 8: Use cards 1 to 6 once each to make three two-card decks. Can A win more pairs than B, B more than C, and C more than A? Find an arrangement, or explain why there is none.

